

- Admin
- ① Noah OH (+ DM OH reminder)
 - ② PSET next week
 - ③ class discussion: project ??

Unsupervised learning for HMMs.

Forward algorithm

Given q, e (models $p(x, y)$)

What is $p(x)$ (for fixed sent x)?

x_1	x_2	$\dots$	x_n	
?	?	$\dots$	?	$\leftarrow y: \text{unknown}$

Viterbi = gave us $\max_y p(x, y) = (\arg \max_y) p(y|x)$

How is this different?

x_1	x_2	$\dots$	x_n	
y_1^*	y_2^*	$\dots$	y_n^*	$\leftarrow \text{best tag seq } y^*$

$\left. \begin{array}{c} \text{prob } p^* \end{array} \right\}$

also want

$p_r \left(\begin{array}{cccc} x_1 & x_2 & \dots & x_n \\ y_1' & y_2' & \dots & y_n' \end{array} \right)$

$\leftarrow \text{some other, non-zero prob tag seq}$

$\left. \begin{array}{c} p' \\ p'' \end{array} \right\} \xrightarrow{\text{add to}} p^*$

$$p(x) = \sum_y p(x, y) = \sum_y p(y) p(x|y) \prod q(y_i | y_{i-1})$$

$\uparrow$ (other valid) $\uparrow$ (other valid)

Align was: $y_i \in K$

enum over all $y = y_1 \dots y_n$ $|K|$

time? $O(|K|^n \times n)$

DP to the rescue! Assume bigram model
 $p(y) = \prod q(y_i | y_{i-1})$

table $\alpha(k, u) = p(x_1, \dots, x_k \wedge y_k = u)$

English meaning: $\alpha(k, u)$ = prob of generating
partial sentence $x_1, \dots, x_k$
and k -th tag $y_k = u$

FP DP 1: solve final problem?

$$p(x) = \sum_{u \in K} \alpha(n, u) \cdot q(\text{STOP} | u)$$

FP DP 2 recursion?

Base case: $\alpha(0, *) = 1$ ($\alpha(0, u \neq *) = 0$)

Say row $k-1$ completely filled

$$\alpha(k, u) = \sum_{w \in K} \alpha(k-1, w) \cdot q(u | w) \cdot e(x_k | u)$$

prob " $x_1, \dots, x_k$
 $\wedge y_k = u$ "

Contains prob of
 $x_1, \dots, x_{k-1}$
and $y_{k-1} = v$ ← use assumption

$y_k = u$

x_k correct for

Note $\sum$ instead of max.

"Forward algorithm".

Time compl $O(n|K|^2)$ $\ll O(n|K|^n)$

(converts HMM to MM language model)

Backward Algorithm

table $\beta(k, u) = p(x_{k+1}, \dots, x_n | y_k = u)$

English = prob of the partial sentence (suffix)
 $x_{k+1}, \dots, x_n$ given (not and like in α)
that k -th tag is u .

Note $p(x) = \beta(0, *) = \text{prob of } x_1, \dots, x_n \text{ given } y_0 = *$

FP DP 2 construct β recursively.

Base case $\beta(n+1, \text{STOP}) = 1$ $\beta(n+1, u \neq \text{STOP}) = 0$

for $k = n, n-1, \dots, 0$:

Assume $\beta(k+1, w)$ exists $\forall w$

$$\beta(k, u) = \sum_v \beta(k+1, v) q(v/u) \cdot e(x_{k+1}/v)$$

prob of
 $x_{k+1} x_{k+2} \dots x_n$
given $y_k = u$

prob of
 $x_{k+2}, \dots, x_n$
given $y_{k+1} = v$

prob of
this

time
 $O(n|k|^2)$

$$\beta(k, u) = \sum_{y_{k+1}, \dots, y_n} p(\underline{x_{k+1}, \dots, x_n} \wedge \underline{y_{k+1}, \dots, y_n} | y_k = u)$$

$$\begin{aligned}
 &= \sum_{y_{k+1}} \left(\sum_{y_{k+2}, \dots, y_n} p(x_{k+2}, \dots, x_n \wedge y_{k+2}, \dots, y_n | y_{k+1}) \cdot p(y_{k+1} | u) \right) \\
 &= \sum_{y_{k+1}} \beta(k+1, y_{k+1}) \cdot p(y_{k+1} | u) \cdot p(x_{k+1} | y_{k+1})
 \end{aligned}$$

Q1 what is the meaning of $\alpha(k, u) \cdot \beta(k, u)$?

prob of $x_1, \dots, x_k$ and $y_k = u$ $\times$ prob of $x_{k+1}, \dots, x_n$ (stop) given $y_k = u$

$$= p(x_1, \dots, x_n \wedge y_k = u)$$

prob of whole x and k -th tag is u .

(by basic PT)

$$= p(x_1, \dots, x_k \wedge y_k = u) \cdot p(x_{k+1}, \dots, x_n | x_1, \dots, x_k, y_k = u)$$

Whole new way of computing

$$p(x) = \sum_u p(x_1, \dots, x_n \wedge y_k = u) = \sum_u \alpha(k, u) \beta(k, u)$$

Cool! can also compute

$$p(y_k = u | x_1, \dots, x_n) = \frac{p(x_1, \dots, x_n \wedge y_k = u)}{p(x_1, \dots, x_n)}$$

$\gamma(k, u)$ prob of k -th tag = u $\gamma(k, u | x)$

$$= \frac{\alpha(k, u) \beta(k, u)}{2 / n \cdot \text{START}} \leftarrow \text{any other}$$

(advanced)

prob of sentence

equiv to $p(x)$

Q2 what is the meaning of

$$\alpha(k, u) \cdot \beta(k+1, v) \cdot q(v|u) \cdot e(x_{k+1}|v)$$

prob of $x_1, \dots, x_k \wedge y_k = u$ $\wedge$ $y_{k+1} = v$ (given $y_k = u$)
 $\wedge$ x_{k+1} (given $y_{k+1} = v$) $\times$ $x_{k+2}, \dots, x_n$ given $y_{k+1} = v$

$$= \text{prob of } x_1, \dots, x_n \wedge y_k = u \wedge y_{k+1} = v$$

prob of sentence and a specific tag bigram (u, v) at position k

$$p(y_k = u \wedge y_{k+1} = v | x_1, \dots, x_n)$$

$$= \frac{p(x_1, \dots, x_n \wedge y_k = u \wedge y_{k+1} = v)}{p(x_1, \dots, x_n)}$$

$$= \frac{\alpha(k, u) \cdot \beta(k+1, v) \cdot q(v|u) \cdot e(x_k|v)}{\beta(0, \text{START})}$$

$\beta(0, \text{START})$ ← or whatever

prob of ^{tag} bigram (u, v) at k -th position

$$= \sum (k, u, v) \quad \sum "x_i"$$

given $q, e \rightarrow$ compute $\alpha, \beta \rightarrow$ useful quantities
 $p(x)$
 $p(u_k = u | x)$

$$\gamma(k, u) \quad \xi(k, u, v) \quad P(y_k = u \wedge y_{k+1} = v | x) \quad \text{etc}$$

What is $\sum_{k=1}^n \gamma(k, u)$

$$\begin{aligned} & \mathbb{E} [\# \text{ occurrences of } u \text{ in tag seq } (x_1, \dots, x_n)] \\ &= \mathbb{E} \left[\sum_{k=1}^n \chi[y_k = u | x_1, \dots, x_n] \right] \quad \chi[E] = 1 \text{ if } E \text{ true, } 0 \text{ otherwise} \\ &= \sum_{k=1}^n \mathbb{E} [\chi[y_k = u | x_1, \dots, x_n]] \\ &= \sum \text{prob}(y_k = u | x_1, \dots, x_n) \\ &= \sum_{k=1}^n \gamma(k, u) \end{aligned}$$

this is exactly $\mathbb{E}[c(u) | x_1, \dots, x_n]$
(write $\mathbb{E}[c(u)]$ when $x_1, \dots, x_n$ understood from context)

What about $\sum_{k=0}^n \xi(k, u, v)$ (let $k=0$ so u can be START)

$$\begin{aligned} &= \sum_{k=0}^n \mathbb{E} [\chi[y_k = u \wedge y_{k+1} = v | x_1, \dots, x_n]] \\ &= \mathbb{E} [\# \text{ occurrences of tag bigram } (u, v) | x_1, \dots, x_n] \\ &= \mathbb{E}[c(u, v)] \quad !! \end{aligned}$$

Overall (summary thus far)

given q, e can compute:

- α, β (forward + backward algs) $O(n|K|^2)$
- $\forall k, u$, prob $y_k = u \quad \text{=: } \gamma(k, u)$ $+ O(n|K|)$
- $\forall u \quad \mathbb{E}[\# u] = \mathbb{E}[c(u)]$ $+ O(n|K|)$
- $\forall u, v$ prob $y_k = u \wedge y_{k+1} = v$ $+ O(n|K|^2)$
- $\forall u, v \quad \mathbb{E}[c(u, v)]$ $+ O(n|K|^2)$

TOTAL = $O(n|K|^2)$

Baum-Welch algorithm

Unsupervised algo for HMM

Given training data $C = \{s_i\}$ $s_i = x_{i,1}, \dots, x_{i,n_i}$
i.e. no tag segs y

+ given initial guess of q, e
output better estimates q', e'

Intuition: use current q, e to "tag" (in expectation) all the data
then recompute q, e using MLE (using $c(u, v)$ $c(u)$ etc.)

if $C =$ one sentence $x_1, \dots, x_n$

let $\forall u, v$ $\dots$ use q, e to

$$q'(v/u) = \frac{\mathbb{E}[c(u,v)]}{\mathbb{E}[c(u)]}$$

compute both, same as "expectationally" tagging wrt q, e + using new tags to recompute $q(v/u)$

+ $O(n|K|^2)$ to get all $q'(v/u)$

what about $e'(x/u)$?

$$= \frac{\mathbb{E}[c(u \rightsquigarrow x)]}{\mathbb{E}[c(u)]}$$

didn't compute this yet but it's easy

$$\begin{aligned} &= \mathbb{E}[\# \text{ occurrences of } y_k = u \text{ in places } x_k = x] \\ &= \sum_{k: x_k = x} \gamma(k, u) \end{aligned}$$

note, computing $\forall x, u$ + $(O(n|K||V|))$ time

If C has > 1 sentence

$$C = \{S_j\}_{j=1}^m$$

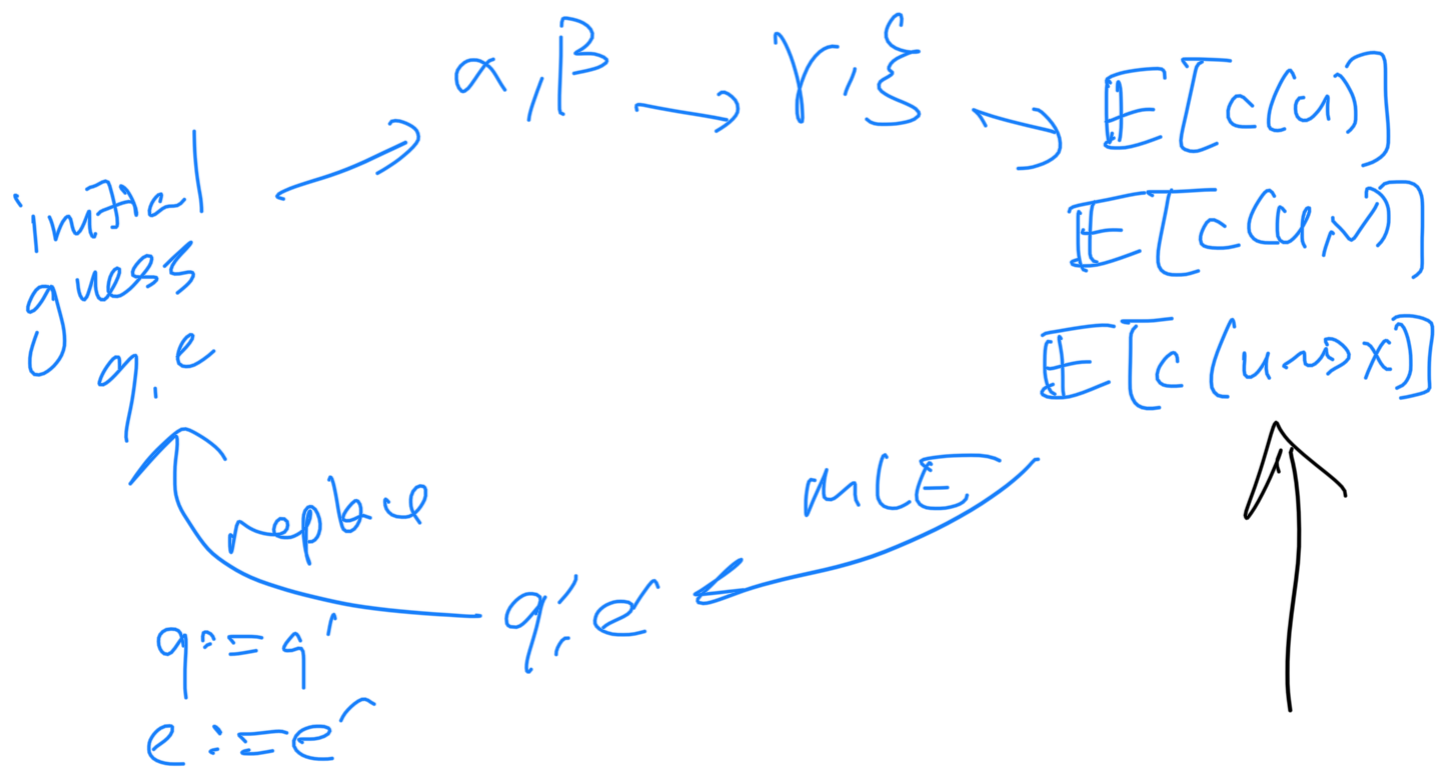
$$q'(v/u) = \frac{\sum_{j=1}^m \mathbb{E}[c(u,v) | S_j]}{\sum_{j=1}^m \mathbb{E}[c(u) | S_j]}$$

ie, use old q, e to do forward backward for each sentence

$$e'(x/u) = \frac{\sum_{j=1}^m \mathbb{E}[c(u \rightsquigarrow x) | S_j]}{\sum_j \mathbb{E}[c(u) | S_j]}$$

Overall: repeat until q, e start to converge / stabilize (not guaranteed in general)

- starting points important ($q^{(0)}, e^{(0)}$)



Expectation

E

Use current guess of params to compute E

Maximization

M

do MLE, maximum likelihood estimate for q', e'
 replace